Customer Product & Profitability Analysis

APL Logistics
Business Analyst Research Paper
Prepared Using Python, Pandas and Streamlit
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Abstract

Discount strategies are widely used by businesses to attract customers, increase sales, and improve market competitiveness. However, excessive discounting can reduce revenue and negatively affect profit margins. This research examines the relationship between discount strategies, revenue, and profitability through a Customer, Product and Profitability Analytics dashboard developed using Python, Pandas, Plotly, and Streamlit.

The dashboard incorporates a discount impact distribution chart and an interactive what-if discount scenario that enables users to evaluate potential financial outcomes under different discount assumptions. In the displayed scenario, an additional discount of 5% produces estimated revenue of $34,945,497.59, estimated profit of $2,579,035.39, and an estimated profit margin of 7.38%.

The analysis demonstrates how interactive dashboards can support financial decision-making by presenting profitability indicators and allowing users to investigate alternative discount scenarios. However, the relationship between the displayed discount assumption and the reported profit increase requires further validation before the results can be used for business decisions.

Keywords: Business Analytics, Discount Strategy, Revenue Analysis, Profitability, Profit Margin, What-If Analysis, Data Visualization, Decision Support Systems.

1. Introduction

Businesses frequently use discounts as a marketing strategy to attract customers, encourage purchases, clear inventory, and increase sales volume. Although discounts may stimulate demand, they also reduce the amount of revenue earned per unit sold. Consequently, businesses must evaluate whether their discount policies generate sufficient benefits to justify the reduction in selling prices.

Traditional financial reporting generally focuses on historical performance, such as total revenue, expenses, and profit. While these indicators are essential, they do not always provide sufficient insight into how alternative pricing decisions might affect future financial outcomes.

Business Analytics addresses this limitation by combining data processing, visualization, and scenario modelling. Interactive dashboards allow decision-makers to examine financial patterns and test hypothetical business conditions without manually recalculating every metric.

This research examines a Customer, Product and Profitability Analytics dashboard designed to investigate discount impact distributions and evaluate an additional discount scenario.

The dashboard provides two principal analytical components:

1. **Discount Impact Distribution:** A chart illustrating the distribution of discount impact ratios across observations.
2. **What-If Discount Scenario:** An interactive control that allows users to adjust the additional discount percentage and observe the resulting estimated revenue, profit, and profit margin.

The study evaluates the usefulness of these components for understanding discount-related financial outcomes and identifies improvements necessary to make the analysis more reliable.

2. Research Problem

Discount decisions involve a trade-off between sales growth and profitability. Although a discount may encourage customers to purchase more products, the reduction in selling price can decrease the profit earned on each transaction.

Without appropriate analytical tools, business managers may struggle to determine whether a proposed discount strategy is financially beneficial.

The central research problem is:

How can interactive Business Analytics dashboards help businesses evaluate the relationship between discount strategies, revenue, and profitability?

The dashboard provides a visual and interactive approach to this problem. However, the reliability of its conclusions depends on the accuracy of the underlying data, the definition of the discount impact ratio, and the mathematical assumptions used in the scenario model.

3. Research Objectives

The primary objective is to examine how an interactive analytics dashboard can support discount-related financial decisions.

The specific objectives are:

1. To examine the distribution of discount impact ratios across the available observations.
2. To evaluate estimated revenue, profit, and profit margin under a selected additional discount scenario.
3. To understand the relationship between discount decisions and business profitability.
4. To demonstrate the usefulness of interactive what-if analysis in financial decision-making.
5. To identify potential inconsistencies in the scenario calculations.
6. To recommend improvements that would make the dashboard more useful for business managers and financial analysts.

4. Research Questions

This study addresses the following questions:

RQ1: What patterns can be observed in the distribution of discount impact ratios?

RQ2: What financial outcomes does the dashboard report when an additional discount of 5% is selected?

RQ3: How can interactive what-if analysis improve the evaluation of alternative discount strategies?

RQ4: What limitations must be addressed before the dashboard can support reliable pricing decisions?

5. Research Methodology

5.1 Research Design

This study uses a quantitative, descriptive, and exploratory analytical approach.

The descriptive component examines the financial indicators and distribution presented in the dashboard. The exploratory component considers how changing the additional discount assumption can help investigate alternative financial scenarios.

The research is based on the displayed dashboard rather than a new analysis of the complete underlying transaction dataset.

Therefore, the numerical results reported in this paper represent the values visible in the supplied dashboard screenshots. They should not be interpreted as independently verified findings from the complete dataset.

5.2 Dashboard Architecture

The dashboard is identified as a Customer, Product and Profitability Analytics application.

According to its displayed attribution, the application was built using the following technologies:

Technology	Purpose
Python	Core programming and analytical calculations
Pandas	Data manipulation and analysis
Plotly	Interactive data visualization
Streamlit	Interactive dashboard interface

These technologies support a workflow in which data can be processed, transformed into analytical measures, visualized, and presented through an interactive interface.

5.3 Analytical Components

The study examines two components of the dashboard.

Component A: Discount Impact Distribution

This chart presents the distribution of discount impact ratios along the horizontal axis and the number of observations along the vertical axis.

The visualization can help identify the concentration of observations around particular discount impact values and reveal less frequent observations at the extremes.

Component B: What-If Discount Scenario

The scenario interface contains a slider labelled Additional Discount (%).

The selected value in the supplied screenshot is 5%.

The dashboard then displays three financial performance indicators:

- Estimated revenue
- Estimated profit
- Estimated profit margin

These indicators provide a summary of the financial outcome calculated by the application under the selected scenario assumptions.

5.4 Analytical Measures

The following measures are relevant to the investigation.

Revenue

Revenue represents the income generated from sales before subtracting operating expenses and other applicable costs.

Profit

Profit represents the difference between revenue and the costs included in the profit calculation.

For a simplified model:

Profit = Revenue − Total Costs

Profit Margin

Profit margin measures the proportion of revenue retained as profit.

Profit Margin (%) = (Profit ÷ Revenue) × 100

Discount Impact Ratio

A discount impact ratio is intended to represent the relationship between a discount and a specified business outcome. Its exact interpretation depends on the formula implemented in the dashboard.

The precise definition and calculation of this ratio should be documented before interpreting positive and negative values.

6. Results and Findings

6.1 Distribution of Discount Impact Ratios

The first dashboard visualization is titled "Distribution of Discount Impact."

The chart shows a pronounced concentration of observations around a discount impact ratio of approximately zero. Smaller groups of observations appear at negative and positive values.

This distribution suggests that discount impact ratios are not evenly distributed across the displayed range.

Several observations can be made.

First, the concentration near zero indicates that many observations have discount impact ratios close to zero, according to the displayed measure.

Second, the smaller bars at the extremes suggest that relatively fewer observations occur at more distant values.

Third, the presence of negative and positive observations indicates variation in the recorded discount impact ratios.

However, the chart alone does not establish why these patterns occur. They could reflect differences in products, customers, transaction characteristics, discount policies, or the mathematical construction of the ratio.

Further analysis would be required to identify the underlying causes.

The chart also appears to contain unusually formatted horizontal-axis labels and abbreviated tooltip values. These should be checked against the original data and visualization code to ensure that the values are displayed correctly.

6.2 Results of the 5% Additional Discount Scenario

The second dashboard visualization presents a what-if scenario with an additional discount of 5%.

The following values are visible in the supplied screenshot.

Performance Indicator	Displayed Value
Additional discount	5%
Estimated revenue	$34,945,497.59
Estimated profit	$2,579,035.39
Estimated profit margin	7.38%
Displayed profit change	+$1,083,256.72

These figures represent the dashboard's displayed scenario output. The underlying calculation has not been independently verified.

6.2.1 Estimated Revenue

The estimated revenue is approximately $34.95 million.

This figure provides an indication of the sales income associated with the selected scenario according to the dashboard's calculation.

Revenue alone, however, is insufficient to establish whether the scenario is financially preferable. The corresponding costs and profit must also be considered.

6.2.2 Estimated Profit

The estimated profit is approximately $2.58 million.

Profit is an important indicator because a pricing strategy should be evaluated according to the financial value it generates, rather than revenue alone.

A discount strategy that increases sales volume may improve overall profit under certain circumstances. However, the increase must be sufficient to compensate for the reduction in selling price and any additional costs associated with fulfilling the increased demand.

The dashboard reports a positive profit change of approximately $1.08 million. This result requires additional investigation because the displayed scenario description states that the additional discount reduces revenue while total costs remain unchanged.

Under a model in which the quantity sold and costs remain fixed, increasing the discount would normally reduce profit rather than increase it.

The positive change could reflect a different calculation, an additional assumption, a comparison against a particular baseline, or an error in the scenario implementation.

Consequently, the reported profit increase should be treated as an unverified dashboard output until the underlying formula is examined.

6.2.3 Estimated Profit Margin

The dashboard reports an estimated profit margin of 7.38%.

Using the displayed revenue and profit figures, the calculation is:

Profit Margin = ($2,579,035.39 ÷ $34,945,497.59) × 100

Estimated Profit Margin ≈ 7.38%

This calculation is consistent with the displayed result.

A profit margin of 7.38% means that approximately $7.38 in profit is generated for every $100 in revenue under the displayed scenario, assuming the dashboard's profit and revenue definitions are consistent.

Whether this margin is satisfactory depends on the business's industry, operating costs, competitive conditions, and financial objectives.

A single margin value cannot establish whether the proposed discount is optimal. It should be compared with the baseline margin and the results of alternative discount scenarios.

6.3 Interpretation of the Combined Results

The two visualizations provide complementary information.

The distribution chart describes variation in the discount impact ratio, while the scenario interface provides a financial summary under a selected additional discount assumption.

Together, they illustrate how descriptive analytics and scenario modelling can be incorporated into a single decision-support interface.

The distribution chart may help analysts identify patterns that deserve further investigation. The what-if component may help managers evaluate how alternative discount assumptions affect financial indicators.

However, these components answer different questions.

A distribution chart describes observed values; it does not establish the causal effect of discounts on profitability. A what-if model calculates outcomes under specified assumptions; it does not automatically predict how customers will respond to a price change.

Reliable business conclusions therefore require a clear distinction between observed data, calculated scenario outputs, and empirically validated predictions.

7. Discussion

7.1 Importance of Discount Analysis

Discount management is an important aspect of pricing and profitability analysis.

A discount may increase demand, improve inventory turnover, or attract new customers. Conversely, it may reduce the contribution earned on existing purchases, particularly when customers would have purchased the products at the original prices.

Businesses must therefore assess both the direct financial impact of price reductions and any behavioural changes that result from them.

The dashboard provides a useful starting point by making discount-related information accessible through visualizations and interactive financial indicators.

7.2 Value of Interactive What-If Analysis

One of the dashboard's principal strengths is its interactive scenario control.

Rather than examining only one fixed report, a decision-maker can adjust the additional discount percentage and observe the corresponding calculated indicators.

This approach can support questions such as:

- What happens to estimated revenue when the additional discount changes?
- How does the profit margin vary across discount scenarios?
- At what discount level does estimated profit reach its maximum?
- Which discount levels maintain a minimum acceptable profit margin?

The dashboard can become more useful if it allows users to compare multiple scenarios side by side.

For example, analysts could compare no additional discount with additional discounts of 5%, 10%, and 15%, provided these values are supported by the model and the calculations are validated.

Such comparisons would make the trade-offs between revenue and profitability easier to evaluate.

7.3 The Importance of Cost and Demand Assumptions

The usefulness of a what-if model depends heavily on its assumptions.

If the model assumes that demand and costs remain constant, the financial effect of a discount is relatively straightforward to calculate.

However, actual customers may respond to lower prices by purchasing more units, switching products, or making purchases they would not otherwise have made.

A more realistic model should therefore distinguish between:

1. The direct reduction in revenue per unit caused by a discount.
2. The potential change in sales volume.
3. The variable costs associated with additional sales.
4. Any changes in fixed costs, marketing costs, or fulfilment expenses.
5. Differences in customer and product profitability.

Without these considerations, a scenario may be mathematically consistent but insufficiently realistic for operational decisions.

7.4 Validating the Reported Profit Increase

The reported increase of $1,083,256.72 is an important issue for further analysis.

The dashboard's explanatory text states that the additional discount reduces revenue while total costs remain unchanged. Under these assumptions, profit should decrease if revenue decreases.

Therefore, the positive displayed profit change warrants investigation.

The following checks are recommended:

- Confirm the baseline profit used for comparison.
- Verify how the additional discount is applied to the original selling price.
- Determine whether sales quantity changes when the discount changes.
- Check whether total costs remain constant or are recalculated.
- Verify whether the profit change is calculated relative to the correct baseline.
- Confirm that the displayed arrow and percentage changes correspond to the underlying calculations.

This validation is essential because a visually attractive dashboard can still communicate misleading conclusions if the financial calculations are incorrect.

8. Recommendations for Dashboard Improvement

Based on the displayed dashboard, the following improvements are recommended.

8.1 Introduce a Baseline Comparison

The dashboard should show baseline financial results alongside scenario results.

Recommended indicators include:

- Baseline revenue
- Scenario revenue
- Revenue change in dollars
- Revenue change as a percentage
- Baseline profit
- Scenario profit
- Profit change in dollars
- Profit change as a percentage
- Baseline profit margin

- Scenario profit margin
- Change in profit margin in percentage points

This would help users distinguish absolute financial values from changes relative to the baseline.

8.2 Validate the Profit Calculation

The profit calculation should be explicitly defined and tested.

For example:

Profit = Revenue − Total Costs

If costs are fixed, a reduction in revenue must result in a corresponding reduction in profit.

If sales volume changes in response to a discount, the model should calculate the resulting revenue and variable costs consistently.

Automated tests should verify that the scenario results behave as expected when the discount changes.

8.3 Add Product-Level and Customer-Level Analysis

Aggregated financial results can conceal substantial differences between products and customers.

The dashboard could be expanded to identify:

- Products with high revenue but low profit margins.
- Products that remain profitable under higher discounts.
- Customer segments with different purchasing responses.
- Transactions where discounts exceed predefined limits.
- Product categories contributing most to total profit.

Such analysis would help businesses move from broad discount policies toward more targeted pricing decisions.

8.4 Introduce Scenario Comparison Visualizations

A line chart showing discount percentage against estimated profit would allow users to evaluate several discount levels simultaneously.

A second chart could show discount percentage against estimated profit margin.

Together, these charts could help identify the discount levels associated with the highest estimated profit and the minimum acceptable margin.

The model should not assume that the highest revenue scenario necessarily produces the highest profit.

8.5 Improve Chart Labels and Data Validation

The discount impact distribution should use clearly formatted axis labels, meaningful units, and a documented definition of the discount impact ratio.

The underlying data should also be checked for:

- Missing values.
- Unexpected negative values.
- Outliers.
- Duplicate observations.
- Incorrect percentage scaling.
- Inconsistent currency units.

These checks would improve interpretability and reduce the risk of drawing conclusions from incorrectly displayed values.

9. Limitations of the Study

This research has several limitations.

First, limited data access: The analysis is based on screenshots of the dashboard rather than the complete underlying dataset. Consequently, the observations, distributions, and financial calculations cannot be independently reproduced.

Second, uncertain metric definitions: The exact formula used to calculate the discount impact ratio is not visible. Its business interpretation therefore remains provisional.

Third, scenario assumptions: The dashboard states that costs remain unchanged, but the underlying calculation has not been inspected. It is also unclear whether customer demand or sales quantity changes in response to the additional discount.

Fourth, absence of a verified baseline: The screenshot displays a profit change but does not clearly establish the baseline revenue, baseline profit, or baseline margin used to calculate that change.

Fifth, no causal analysis: The displayed distribution does not establish that discounts cause changes in revenue or profitability. Such conclusions would require a suitable experimental design or a robust statistical analysis.

Sixth, limited generalizability: The findings cannot be generalized to other businesses, industries, or customer populations without additional evidence.

These limitations mean that the dashboard should currently be treated as an exploratory analytical tool rather than a validated forecasting system.

10. Future Research

Future research could extend the dashboard into a more comprehensive pricing and profitability decision-support system.

First, the underlying transaction dataset could be analyzed to investigate the relationship between discount levels, sales volume, revenue, and profit.

Second, statistical methods could be used to identify differences in discount outcomes across product categories and customer segments.

Third, regression analysis could investigate the association between discount percentages and sales performance while accounting for other relevant variables.

Fourth, controlled experiments or carefully designed observational studies could help estimate customer price sensitivity.

Fifth, a profit optimization model could identify the discount level that maximizes expected profit under specified demand and cost assumptions.

Finally, the dashboard could incorporate scenario uncertainty by displaying ranges of possible outcomes rather than a single estimated value.

These extensions would improve the analytical depth of the project and provide a stronger foundation for evidence-based pricing decisions.

11. Conclusion

This research examined the application of Business Analytics to discount strategy evaluation through a Customer, Product and Profitability Analytics dashboard developed using Python, Pandas, Plotly, and Streamlit.

The dashboard combines a discount impact distribution chart with an interactive what-if scenario. In the displayed 5% additional discount scenario, the reported estimated revenue is $34,945,497.59, estimated profit is $2,579,035.39, and estimated profit margin is 7.38%.

The distribution chart provides an initial view of the variation in discount impact ratios, while the scenario interface illustrates how financial indicators can be presented under alternative assumptions.

The analysis highlights the practical value of interactive dashboards for financial exploration and business decision support. It also identifies an important validation requirement: the reported profit increase of $1,083,256.72 must be reconciled with the stated assumption that the discount reduces revenue while costs remain unchanged.

Ultimately, the effectiveness of a profitability dashboard depends not only on its visual presentation but also on the accuracy of its calculations, the transparency of its assumptions, and the quality of the underlying data.

With validated financial formulas, baseline comparisons, product-level analysis, and demand-sensitive scenario modelling, the dashboard could become a more reliable tool for evaluating discount strategies and supporting profitability-focused business decisions.

References

The following references provide a theoretical foundation for further development of the research. They are not evidence that the specific dashboard calculations have been independently validated.

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Appendix A: Dashboard Summary

Project title: Customer, Product and Profitability Analytics

Analytical focus: Discount impact distribution and financial scenario analysis

Tools: Python, Pandas, Plotly, Streamlit

Selected scenario: 5% additional discount

Displayed estimated revenue: $34,945,497.59

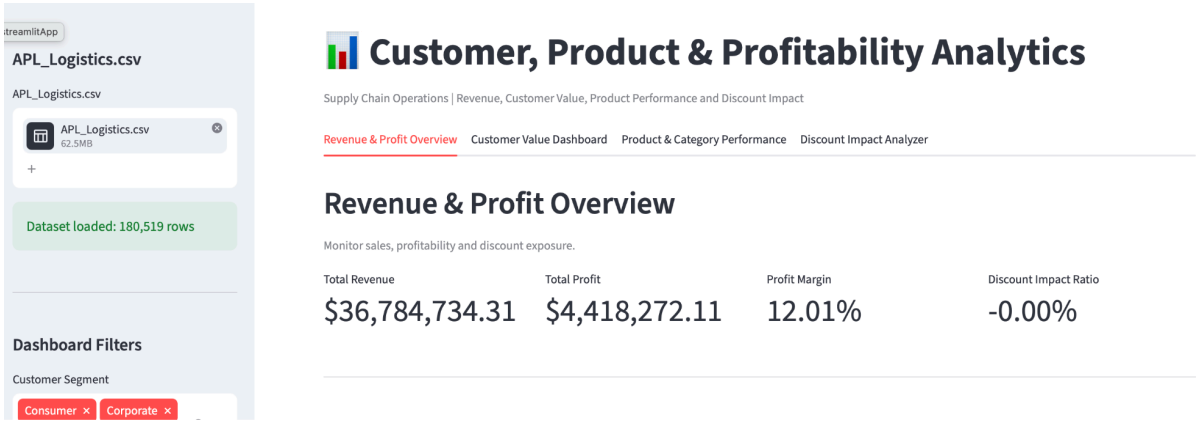
Displayed estimated profit: $2,579,035.39

Displayed estimated profit margin: 7.38%

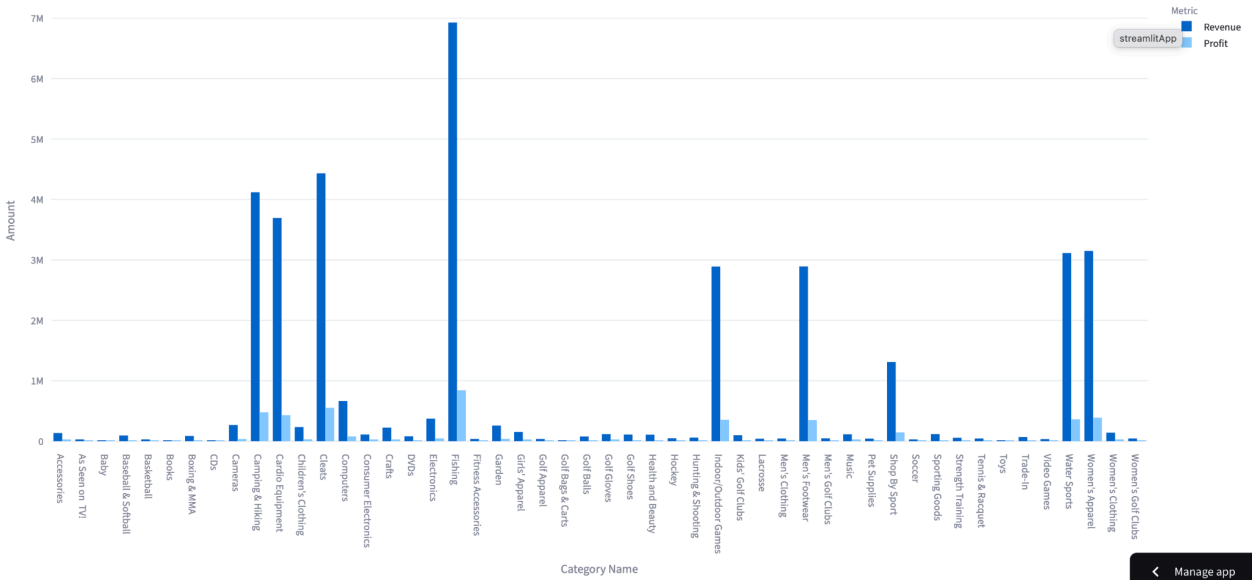
Displayed profit change: +$1,083,256.72

Primary business application: Evaluating alternative discount assumptions and examining their potential financial implications.

Recommended next step: Validate the scenario calculations against the underlying dataset before using the reported profit increase to recommend a discount policy.



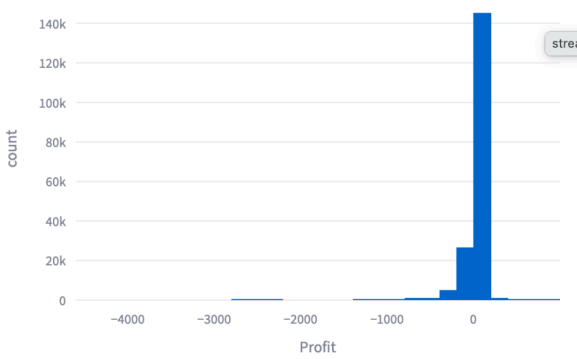
Revenue vs Profit by Category



	Type	Days for shipping (real)	Days for shipment (scheduled)	Benefit per order	Sales per customer	Delivery Status	Late_delivery_risk	Category Id	Category Name	Customer City	Customer Country	Cus
0	DEBIT	6	4	159.69	472.45	Late delivery	1	9	Cardio Equipment	Brownsville	EE. UU.	Rich
1	DEBIT	4	4	48.71	167.96	Shipping on time	0	29	Shop By Sport	Littleton	EE. UU.	Mar
2	DEBIT	4	4	87.36	181.99	Shipping on time	0	48	Water Sports	Littleton	EE. UU.	Mar
3	DEBIT	6	4	-41.89	175.99	Late delivery	1	48	Water Sports	Littleton	EE. UU.	Mar
4	DEBIT	6	4	10	40	Late delivery	1	24	Women's Apparel	Littleton	EE. UU.	Mar
5	DEBIT	6	4	105.63	290.98	Late delivery	1	43	Camping & Hiking	Littleton	EE. UU.	Mar
6	DEBIT	4	4	64.1	142.44	Shipping on time	0	46	Indoor/Outdoor Games	Littleton	EE. UU.	Mar
7	DEBIT	4	4	22.67	83.97	Shipping on time	0	46	Indoor/Outdoor Games	Littleton	EE. UU.	Mar
8	DEBIT	4	4	3.52	351.98	Shipping on time	0	45	Fishing	Littleton	EE. UU.	Mar
9	DEBIT	4	4	40.34	98.38	Shipping on time	0	17	Cleats	Littleton	EE. UU.	Mar
10	PAYMEN	3	4	-131.7	84.97	Advance shipping	0	46	Indoor/Outdoor Games	Littleton	EE. UU.	Mar
11	DEBIT	3	4	34.5	74.99	Advance shipping	0	9	Cardio Equipment	Caguas	Puerto Rico	Anr
12	DEBIT	3	4	10.8	40	Advance shipping	0	24	Women's Apparel	Caguas	Puerto Rico	Anr
13	DEBIT	4	4	-42.61	24.35	Shipping on time	0	13	Electronics	Caguas	Puerto Rico	Anr
14	DEBIT	3	4	27.41	94.5	Advance shipping	0	24	Women's Apparel	Caguas	Puerto Rico	Anr
15	DEBIT	0	0	-140.73	191.99	Shipping on time	0	48	Water Sports	Caguas	Puerto Rico	Anr
16	DEBIT	0	0	11.4	189.99	Shipping on time	0	48	Water Sports	Caguas	Puerto Rico	Anr
17	DEBIT	0	0	46.51	98.96	Shipping on time	0	46	Indoor/Outdoor Games	Caguas	Puerto Rico	Anr
18	DEBIT	4	4	-274.39	391.98	Shipping on time	0	45	Fishing	Caguas	Puerto Rico	Anr
19	TRANSF	4	4	83.69	269.98	Shipping on time	0	43	Camping & Hiking	Caguas	Puerto Rico	Anr

Profit Distribution

Distribution of Order-Level Profit



Profit Margin Distribution

Distribution of Profit Margin



streamlitApp

Customer Value Dashboard

Customers / Segments

3

Average Customer Profit

$1,472,757.37

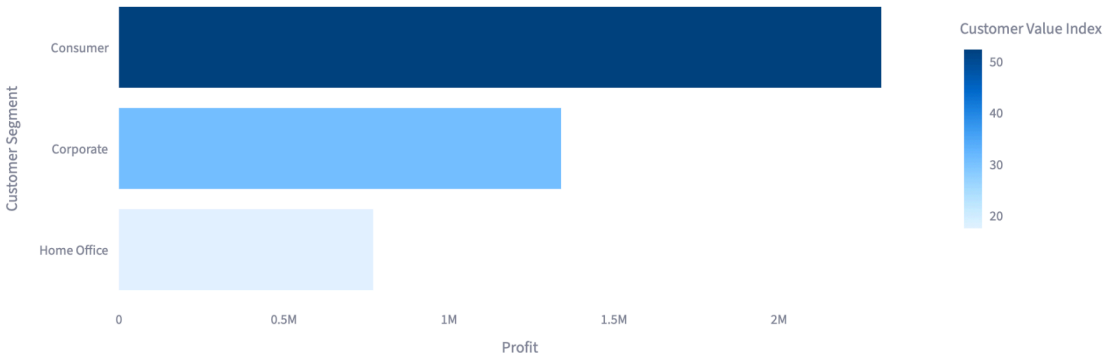
Highest Customer Profit

$2,309,043.90

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Top Customers by Profit

Top 10 Customers by Profit

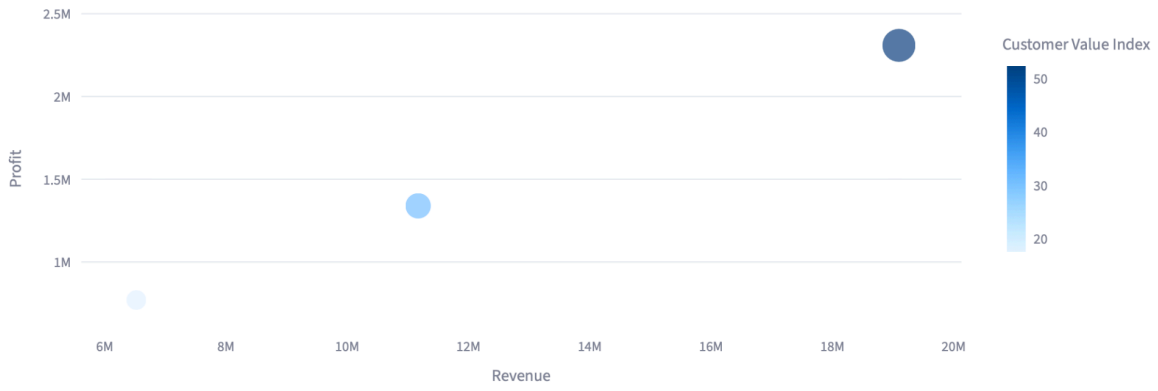


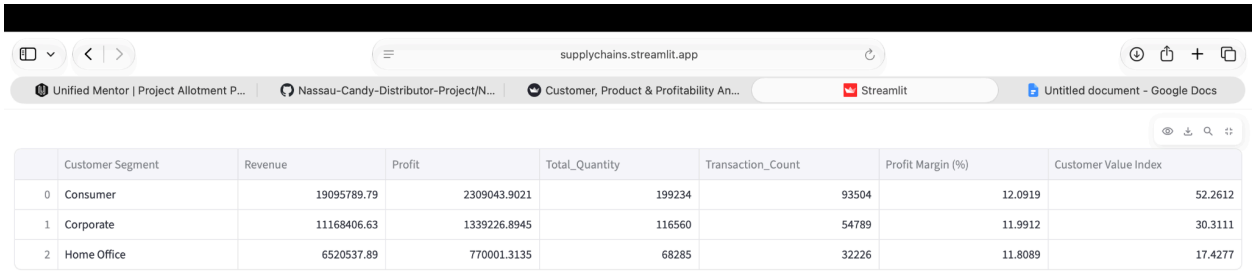
Lowest Customers by Profit

	Customer Segment	Revenue	Profit	Total_Quantity	Transaction_Count	Profit Margin (%)	Customer Value Index
2	Home Office	6520537.89	770001.3135	68285	32226	11.8089	17.4277
1	Corporate	11168406.63	1339226.8945	116560	54789	11.9912	30.3111
0	Consumer	19095789.79	2309043.9021	199234	93504	12.0919	52.2612

Customer Profit Contribution

Customer Revenue vs Profit

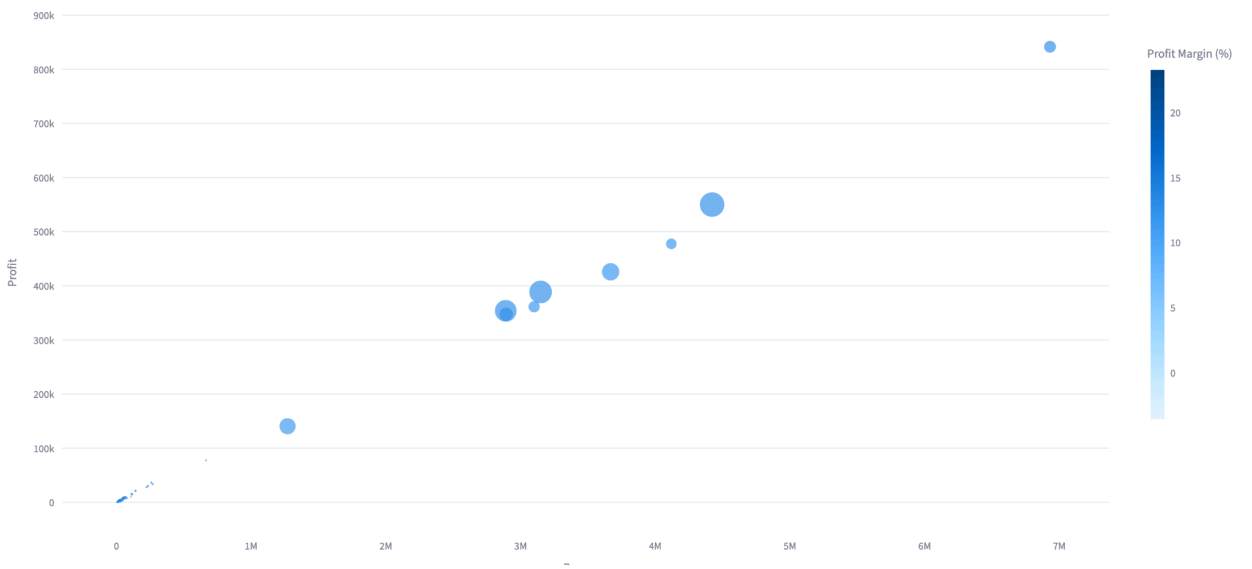




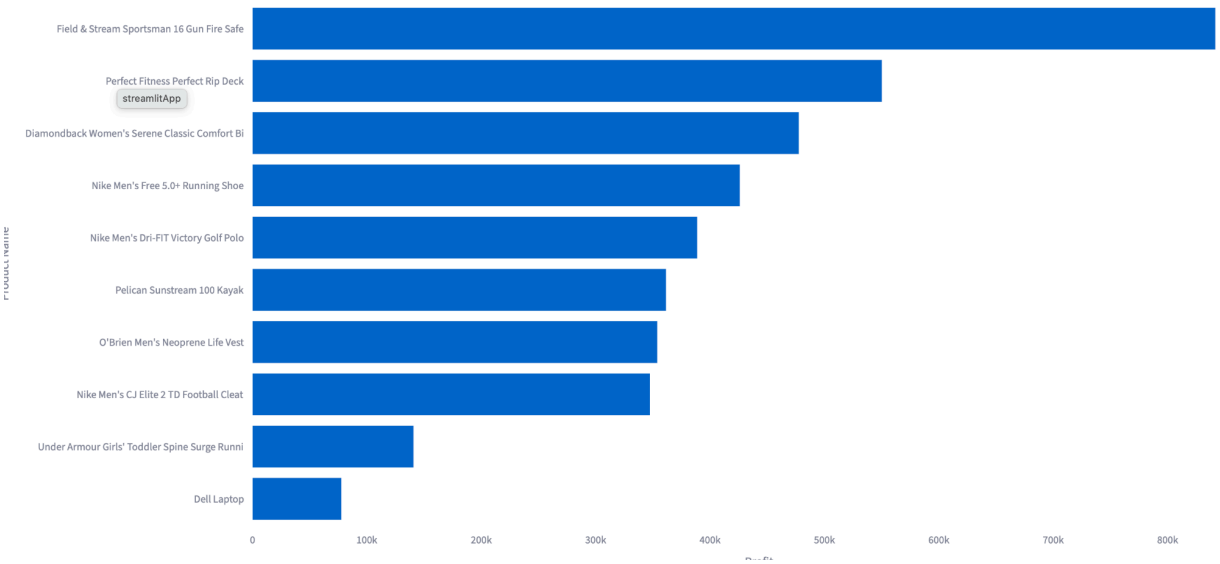
	Customer Segment	Revenue	Profit	Total_Quantity	Transaction_Count	Profit Margin (%)	Customer Value Index
0	Consumer	19095789.79	2309043.9021	199234	93504	12.0919	52.2612
1	Corporate	11168406.63	1339226.8945	116560	54789	11.9912	30.3111
2	Home Office	6520537.89	770001.3135	68285	32226	11.8089	17.4277

streamlitApp

Product Revenue vs Profitability

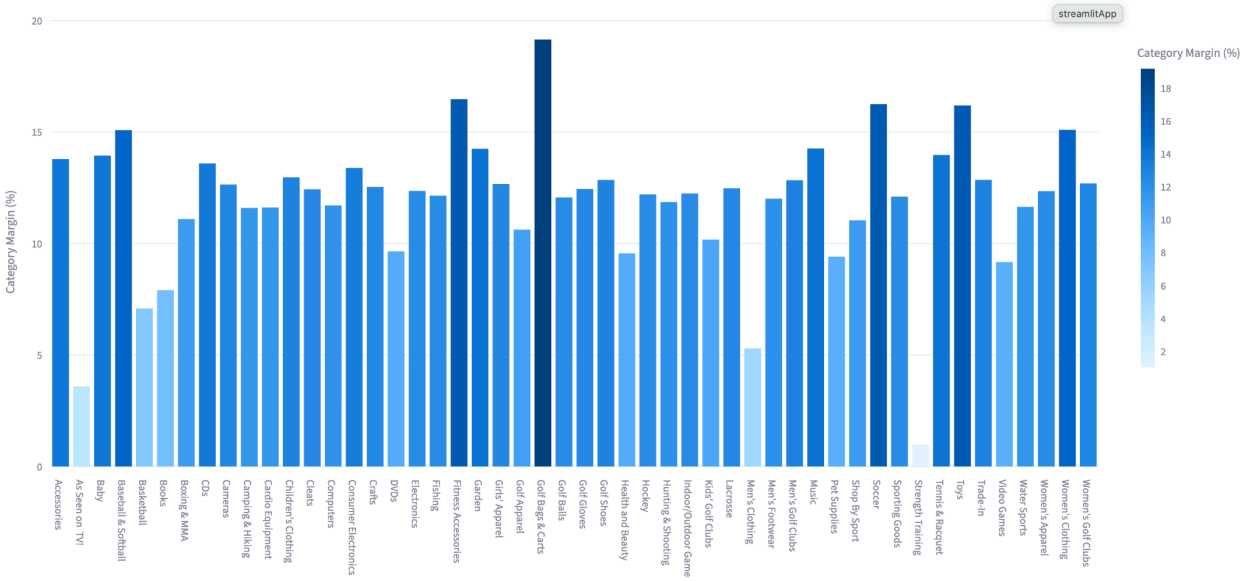


Top 10 Products by Profit



	Product Name	Revenue	Profit	Quantity	Profit Margin (%)	Profit per Unit
70	Pelican Sunstream 100 Kayak	3099845	361361.931	15500	11.6574	23.3137
67	O'Brien Men's Neoprene Life Vest	2888993.94	353791.9266	57803	12.2462	6.1206
56	Nike Men's CJ Elite 2 TD Football Cleat	2891757.54	347367.0774	22246	12.0123	15.6148
102	Under Armour Girls' Toddler Spine Surge Runni	1269082.65	140687.2194	31735	11.0857	4.4332
18	Dell Laptop	663000	77610	442	11.7059	175.5882
46	Lawn mower	257768.72	36737.3684	484	14.2521	75.9037
110	Web Camera	267607.68	33844.2348	592	12.647	57.1693
11	Children's heaters	232829.2	30189.234	652	12.9663	46.3025
74	Porcelain crafts	223356.32	28007.2212	484	12.5393	57.8662
81	Summer dresses	140283	21182.733	650	15.1	32.5888
75	Rock music	113122.1	16136.8415	434	14.265	37.1817
41	Industrial consumer electronics	108991.28	14591.176	431	13.3875	33.8542
78	Smart watch	117006.75	14165.355	357	12.1064	39.6789
26	First aid kit	106080.48	10139.184	362	9.558	28.0088
116	adidas Youth Germany Black/Red Away Match Soc	67830	9128	969	13.4572	9.42
114	adidas Men's F10 Messi TRX FG Soccer Cleat	56330.61	9050.0914	939	16.066	9.638
17	DVDs	79395.54	7660.108	483	9.648	15.8594
97	Titleist Pro V1x High Numbers Personalized Go	47206.92	7560.3858	908	16.0154	8.3264
44	LIJA Women's Eyelet Sleeveless Golf Polo	61490	7485.4	946	12.1734	7.9127
94	Titleist Pro V1 High Numbers Personalized Gol	44243.49	6819.5283	851	15.4136	8.0135

Profit Margin by Product Category



	Category Name	Revenue	Profit	Category Margin (%)
23	Golf Bags & Carts	10369.39	1985.4832	19.1475
19	Fitness Accessories	35601.44	5865.8498	16.4764
39	Soccer	26477.05	4303.195	16.2525
43	Toys	6104.66	988.5164	16.1928
48	Women's Clothing	140283	21182.733	15.1
3	Baseball & Softball	94057.15	14191.2297	15.0879
36	Music	113122.1	16136.8415	14.265
20	Garden	257768.72	36737.3684	14.2521
42	Tennis & Racquet	44585.09	6230.6651	13.9748
2	Baby	12229.56	1705.0488	13.942
0	Accessories	133671.51	18429.3753	13.7871
7	CDs	3059.59	415.8107	13.5904
14	Consumer Electronics	108991.28	14591.176	13.3875
11	Children's Clothing	232829.2	30189.234	12.9663
44	Trade-In	68721.78	8835.2475	12.8565
26	Golf Shoes	107998	13880.08	12.8522
35	Men's Golf Clubs	47035.8	6038.322	12.8377
49	Women's Golf Clubs	44545.97	5654.4519	12.6935
21	Girls' Apparel	151706.2	[illegible]424	12.6716
8	Cameras	267607.68	33844.2348	12.647

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Discount Impact Analyzer

Compare original item value, recorded sales and estimated discount exposure.

Original Item Value

$36,784,734.31

Recorded Sales

$36,784,734.31

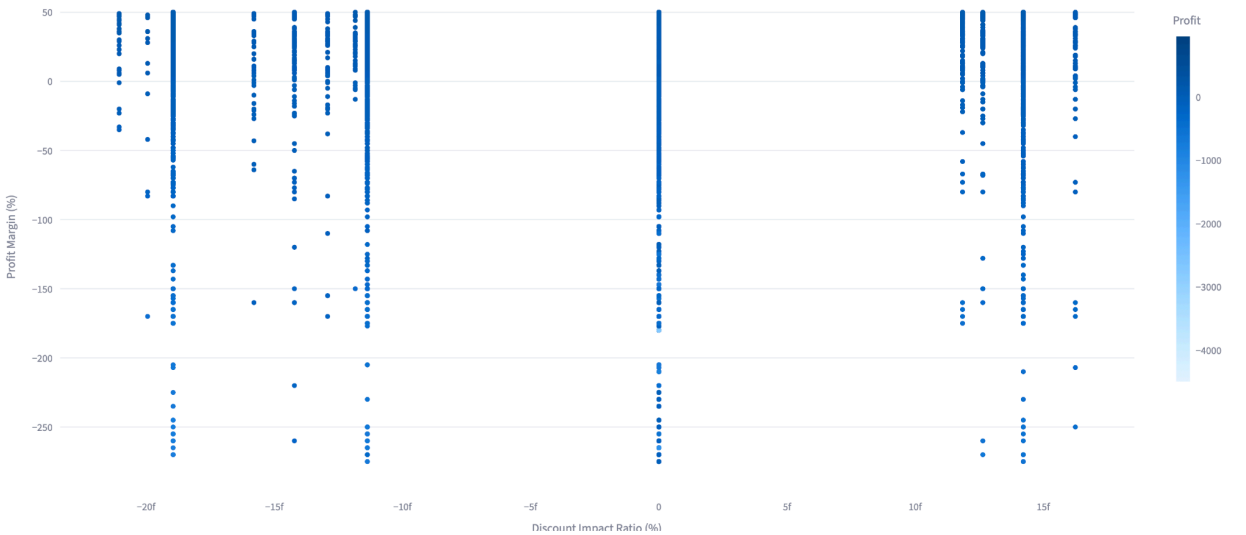
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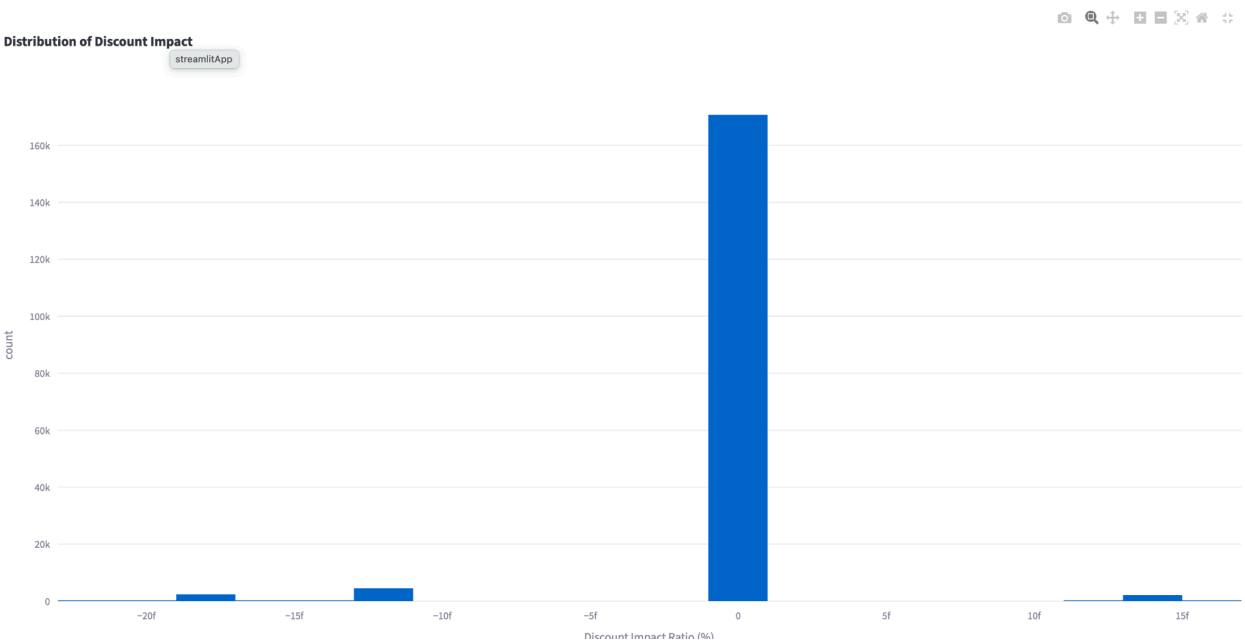
[illegible]ount Amount

$-0.00

Discount Impact vs Profit Margin

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What-If Discount Scenario

Additional Discount (%)

5

Estimated Revenue

$34,945,497.59

Estimated Profit

$2,579,035.39

↑ $-1,839,236.72

Estimated Profit Margin

7.38%

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Scenario assumption: the additional discount reduces revenue while total costs remain unchanged. This is a simplified illustration, not a forecast of actual customer demand or cost behavior.